

# Mark Scheme (Results) January 2007

GCE

GCE Mathematics

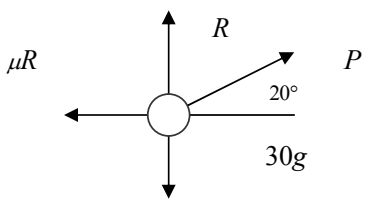
Mechanics M1 (6677)



**January 2007**  
**6677 Mechanics M1**  
**Mark Scheme**

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks                                                                                                                                   |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 1.              | <p>(a) <math>P \sin 30^\circ = 24</math><br/> <math>P = 48</math></p> <p>(b) <math>Q = P \cos 30^\circ</math><br/> <math>\approx 41.6</math>      accept <math>24\sqrt{3}</math>, awrt 42</p>                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>M1 A1<br/> A1    <u>3</u></p> <p>M1 A1<br/> A1    <u>3</u>    <b>6</b></p>                                                           |
| 2.              | <p>(a) <math>M(C) \quad 80 \times x = 120 \times 0.5</math><br/> <math>x = 0.75</math> *      cso</p> <p>(b) Using reaction at <math>C = 0</math><br/> <math>M(D) \quad 120 \times 0.25 = W \times 1.25</math>      ft their <math>x</math><br/> <math>W = 24 \quad (\text{N})</math></p> <p>(c) i      <math>X = 24 + 120 = 144 \quad (\text{N})</math>      ft their <math>W</math></p> <p>(d) The weight of the rock acts precisely at <math>B</math>.</p>                                                                                                                                                    | <p>M1 A1<br/> A1    <u>3</u></p> <p>B1<br/> M1 A1<br/> A1    <u>4</u></p> <p>M1 A1ft<br/> <u>2</u><br/> B1    <u>1</u>    <b>10</b></p> |
| 3.              | <p>(a) <math>\mathbf{a} = \frac{(15\mathbf{i} - 4\mathbf{j}) - (3\mathbf{i} + 2\mathbf{j})}{4} = 3\mathbf{i} - 1.5\mathbf{j}</math></p> <p>(b) N2L      <math>\mathbf{F} = m\mathbf{a} = 6\mathbf{i} - 3\mathbf{j}</math>      ft their <math>\mathbf{a}</math><br/> <math> \mathbf{F}  = \sqrt{6^2 + 3^2} \approx 6.71 \quad (\text{N})</math>      accept <math>\sqrt{45}</math>, awrt 6.7</p> <p>(c) <math>\mathbf{v}_6 = (3\mathbf{i} + 2\mathbf{j}) + (3\mathbf{i} - 1.5\mathbf{j})6</math>      ft their <math>\mathbf{a}</math><br/> <math>= 21\mathbf{i} - 7\mathbf{j} \quad (\text{ms}^{-1})</math></p> | <p>M1 A1 <u>2</u></p> <p>M1 A1<br/> M1 A1 <u>4</u></p> <p>M1 A1ft<br/> A1    <u>1</u>    <b>9</b></p>                                   |

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks                                                                                                   |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 4.              | <p>(a) CLM <math>0.3u = 0.3 \times (-2) + 0.6 \times 5</math><br/><math>u = 8</math></p> <p>(b) <math>I = 0.6 \times 5 = 3</math> (Ns)</p> <p>(c) <math>v = u + at \Rightarrow 5 = a \times 1.5</math> (<math>a = \frac{10}{3}</math>)<br/>N2L <math>R = 0.6 \times \frac{10}{3} = 2</math></p>                                                                                                                                                                          | <p>M1 A1<br/>M1 A1 <u>4</u></p> <p>M1 A1 <u>2</u></p> <p>M1 A1<br/>M1 A1 <u>4</u> <b>10</b></p>         |
| 5.              | <p>(a) <math>v^2 = u^2 + 2as \Rightarrow 0^2 = 21^2 - 2 \times 9.8 \times h</math><br/><math>h = 22.5</math> (m)</p> <p>(b) <math>v^2 = u^2 + 2as \Rightarrow v^2 = 0^2 + 2 \times 9.8 \times 24</math> or equivalent<br/>(= 470.4)<br/><math>v \approx 22</math> (<math>\text{ms}^{-1}</math>) accept 21.7</p> <p>(c) <math>v = u + at \Rightarrow -\sqrt{470.4} = 21 - 9.8t</math> or equivalent<br/>– 1 each error<br/><math>t \approx 4.4</math> (s) accept 4.36</p> | <p>M1 A1<br/>A1 <u>3</u></p> <p>M1 A1<br/>A1 <u>3</u></p> <p>M1 A2 (1, 0)<br/>A1 <u>4</u> <b>10</b></p> |

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks                                                                                                                                       |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 6.              | <p>(a)</p>  <p>Use of <math>F = \mu R</math></p> <p><math>P \cos 20^\circ = \mu R</math></p> <p>i <math>R + P \sin 20^\circ = 30g</math></p> <p><math>P \cos 20^\circ = \mu(30g - P \sin 20^\circ)</math></p> <p><math>P = \frac{0.4 \times 30g}{\cos 20^\circ + 0.4 \sin 20^\circ}</math></p> <p><math>\approx 110 \text{ (N)}</math> accept 109</p> <p>(b)</p> <p>i <math>R + 150 \sin 20^\circ = 30g</math></p> <p><math>(R \approx 242.7)</math></p> <p>N2L <math>\quad \quad \quad 150 \cos 20^\circ - \mu R = 30a</math></p> <p><math>a \approx \frac{150 \cos 20^\circ - 0.4 \times 242.7}{30}</math></p> <p><math>= 1.5 \text{ (ms}^{-2}\text{)}</math> accept 1.46</p> | <p>B1</p> <p>M1 A1</p> <p>M1 A1</p> <p>M1</p> <p>M1</p> <p>A1 <u>8</u></p> <p>M1 A1</p> <p>M1 A1</p> <p>M1</p> <p>A1 <u>6</u> <b>14</b></p> |

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks                                                                                                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.              | <p>(a) N2L <math>Q</math> <math>2g - T = 2a</math><br/> N2L <math>P</math> <math>T - 3g \sin 30^\circ = 3a</math></p> <p>(b) <math>2g - 3g \sin 30^\circ = 5a</math><br/> <math>a = 0.98 \text{ (ms}^{-2}\text{)} \star</math> cso</p> <p>(c) <math>T = 2(g - a)</math> or equivalent<br/> <math>\approx 18 \text{ (N)}</math> accept 17.6</p> <p>(d) The (magnitudes of the) accelerations of <math>P</math> and <math>Q</math> are equal</p> <p>(e) <math>v^2 = u^2 + 2as \Rightarrow v^2 = 2 \times 0.98 \times 0.8 (=1.568)</math><br/> <math>v \approx 1.3 \text{ (ms}^{-1}\text{)}</math> accept 1.25</p> <p>(f) N2L for <math>P</math> <math>-3g \sin 30^\circ = 3a</math><br/> <math>a = (-)\frac{1}{2}g</math><br/> <math>s = ut + \frac{1}{2}at^2 \Rightarrow 0 = \sqrt{1.568}t - \frac{1}{2}4.9t^2</math> or equivalent<br/> <math>t = 0.51 \text{ (s)}</math> accept 0.511</p> <p><i>A maximum of one mark can be lost for giving too great accuracy.</i></p> | <p>M1 A1<br/> M1 A1 <u>4</u></p> <p>M1<br/> A1 <u>2</u></p> <p>M1<br/> A1 <u>2</u></p> <p>B1 <u>1</u></p> <p>M1<br/> A1 <u>2</u></p> <p>M1 A1<br/> M1 A1<br/> A1 <u>5</u> <b>16</b></p> |